

AI TRAINING – WEEK 2

ASSIGNMENT 1

Lists, Tuples & Sets

Learn → Practice → Build → Explain

1. What You Will Learn

You will learn three important Python collection types: lists, tuples, and sets. You will practice accessing and modifying them, identify common errors, and build two small real-world utilities.

- Lists – indexing, slicing, methods, and mutability
- Tuples – indexing, slicing, methods, and immutability
- Sets – unique values and common set operations
- Difference between list, tuple, and set
- Common errors and how to fix them
- Build a simple shopping cart
- Remove duplicate values using a set

2. Student Deliverable – Keep It Simple

Submit ONE Jupyter Notebook only.

File name: Week2_Day1_Lists_Tuples_Sets_YourName.ipynb

Your notebook should contain:

- Part A – Basic practice
- Part B – Lists
- Part C – Tuples
- Part D – Sets
- Part E – Debugging
- Part F – Shopping Cart mini project
- Part G – Duplicate Removal utility
- Part H – Explain what you learned

No separate Python files, Word/PDF files, or screenshots are required.

3. Part A – Understand the Basics

Create and print:

```
fruits = ["apple", "banana", "orange"]  
coordinates = (10, 20)  
numbers = {1, 2, 3, 4}
```

```
print(fruits)
```

```
print(coordinates)
```

```
print(numbers)
```

Output:

```
['apple', 'banana', 'orange']
```

```
(10, 20)
```

```
{1, 2, 3, 4}
```

What is a list?

Ans: A list is a collection of value stored in square brackets []

lists are ordered and changeable

- What is a tuple?

Ans: Tuple is a collection of values stored in parentheses()

Tuples are ordered but cannot be changed

- What is a set?

Ans: A set is a collection of unique values stored in curly brackets {}

sets do not allow duplicate values

- Which one allows duplicate values?

Ans: list and Tuple allow duplicate values

- Which one is mutable?

Ans : list is the mutable

4. Part B – List Practice

```
fruits = ["apple", "banana", "orange", "mango", "grapes"]
```

- Print the first item.

Ans: print("banana")

- Print the last item.

Ans: print("grapes")

- Print the first three items using slicing.

Ans:fruits=["apple","banana","orange","mango","grapes"]

```
print(fruits[:3])
```

Output:['apple', 'banana', 'orange']

- Change "banana" to "watermelon".

Ans:fruits=["apple","banana","orange","mango","grapes"]

```
fruits[1]="watermelon"
```

```
print(fruits)
```

Output:['apple', 'watermelon', 'orange', 'mango', 'grapes']

- Add "kiwi".

Ans:fruits = ["apple","banana","orange"]

```
fruits.append("kiwi")
```

```
print(fruits)
```

Output:['apple', 'banana', 'orange', 'kiwi']

- Remove "orange".

Ans:fruits = ["apple","banana","orange"]

```
fruits.remove("orange")
```

```
print(fruits)
```

Output:['apple', 'banana']

- Print the final list.

Ans:fruits = ["apple", "banana", "orange", "mango", "grapes"]

```
fruits.append("kiwi")
```

```
print(fruits)
```

Output:['apple', 'banana', 'orange', 'mango', 'grapes', 'kiwi']

- Find the length of the list.

Ans:fruits = ["apple","banana","orange","kiwi"]

```
print(len(fruits))
```

Output:4

Practice: append(), insert(), remove(), pop(), sort(), reverse(), len()

5. Part C – Tuple Practice

```
student = ("Phani", 34, "Hyderabad", "Python")
```

- Print the first item.

```
Ans:print(student[0])
```

- Print the last item.

```
Ans:print(student[-1])
```

- Print the first two items using slicing.

```
Ans:print(student[:2])
```

- Find the length.

```
Ans:print(len(student))
```

- Try changing 34 to 35.

```
Ans:student[1] = 35  
print(student[1])
```

- Observe the error and explain why it happened.

Ans:A tuple is immutable which means its items cannot be changed after the tuple is created

Remember: tuples are generally used when the collection should not be changed.

6. Part D – Set Practice

```
numbers = {10, 20, 30, 20, 10, 40}
```

- Print the set.

```
Ans:print(numbers)
```

- Observe what happened to duplicates.

Ans:set contains only unique values

duplicates values delete automatically

- Add 50.

```
Ans:numbers.add(50)
```

- Remove 20.

```
Ans:numbers.remove(20)
```

- Check whether 30 exists.

```
Ans:print(30 in numbers)
```

- Find the number of unique values.

```
Ans:print(len(set(numbers)))
```

Important: sets are unordered collections. Do not depend on a particular display order.

7. Part E – List vs Tuple vs Set

Feature	List	Tuple	Set
Ordered?	Yes	Yes	No
Allows duplicates?	Yes	Yes	No
Mutable?	Yes	No	Yes
Supports indexing?	Yes	Yes	No
Main use?	Store changeable collection	Store unchangeable collection	Store unique values

8. Part F – Quick Debugging

Find the error, fix it, and explain the mistake.

Debugging Problem 1

```
fruits = ["apple", "banana", "orange"]
print(fruits[5])
```

```
code:fruits = ["apple", "banana","orange"]
```

```
print(fruits[2])
```

Output:orange

Explanation:Trying to access an index that is outside the list range causes an IndexError

Debugging Problem 2

```
numbers = (10, 20, 30)
numbers[0] = 100
```

```
code:numbers = [10,20,30]
```

```
    numbers[0] =100
```

```
    print(numbers)
```

output:[100,20,30]

Exaplination:trying to change a tuple causes a typeError

Debugging Problem 3

```
items = {1, 2, 3}
print(items[0])
```

```
code: items = {1,2,3}
```

```
    print(1 in items)
```

Output:True

Explanation: A set does not support indexing.

Debugging Problem 4

```
fruits = ["apple", "banana"]
fruits.add("orange")

code: fruits = ["apple", "banana"]

    fruits.append("orange")

    print(fruits)
```

Output: ['apple', 'banana', 'orange']

Explanation:

The mistake is using `add()` with a list. Lists use `append()` to add an item.

9. Part G – Hands-on Activity 1: Shopping Cart

Start with: `cart = ["Rice", "Milk", "Bread"]`

Build a simple shopping cart using a list. Your program should:

- Display the cart.
- Add a new item.
- Remove an item.
- Display the number of items.
- Check whether a particular item exists.
- Display the final cart.

```
code: cart = ["Rice", "Milk", "Bread"]

    def show_cart():
        print("Cart:", cart)

    def add_item(item):
        cart.append(item)

    def remove_item(item):
        cart.remove(item)

    show_cart()

    add_item("Eggs")

    remove_item("Milk")

    print("Number of items:", len(cart))
```

```
print("is Bread in the cart?", "Bread" in cart)

show_cart()

Output: Cart: ['Rice', 'Milk', 'Bread']

Number of items: 3

is Bread in the cart? True

Cart: ['Rice', 'Bread', 'Eggs']

Bonus: create add_item(), remove_item(), and show_cart() functions.
```

10. Part H – Hands-on Activity 2: Duplicate Removal

Create a program that removes duplicate values from a list using a set.

Example input: [10, 20, 10, 30, 20, 40, 30]

Expected result: a collection containing only unique values.

Test with:

- [1, 1, 2, 2, 3, 3]
- ["apple", "banana", "apple", "orange", "banana"]
- [10, 20, 30]

```
def remove_duplicates(items):
    return list(set(items))

print(remove_duplicates([10, 20, 10, 30, 20, 40, 30]))

print(remove_duplicates([1, 1, 2, 2, 3, 3]))

print(remove_duplicates(["apple", "banana", "apple", "orange", "banana"]))

print(remove_duplicates([10, 20, 30]))
```

OUTPUT:

[40, 10, 20, 30]

[1, 2, 3]

['orange', 'apple', 'banana']

[10, 20, 30]

Why is a set useful for this problem?

Answer: Because a set automatically throws away duplicate items for you. You don't have to write any extra code or loops to check for them.

11. Part I – Explain in Your Own Words

1. What is the main difference between a list and a tuple?

Answer: Lists can be changed after you make them, but tuples are locked and cannot be changed.

2. Why can't we normally change an item inside a tuple?

Answer: Because they are designed to be permanent. This keeps the data safe so it doesn't get changed by mistake

3. Why are sets useful for removing duplicates?

Answer: Because a set only allows unique items. It automatically throws away any repeated values.

4. What does indexing mean?

Answer: : It means finding a specific item in a collection by using its exact position number (which starts at 0)

5. What does slicing mean?

Answer: Grabbing a smaller chunk or section of a list or word, instead of just a single item .

6. What does mutable mean?

Answer: It means it can be changed, edited, or updated after you create it.

7. When would you choose a list instead of a tuple?

Answer: When you know you will need to add, remove, or change the items later

12. Student Submission Checklist

- I completed all practice sections.✓
- I tested my code.✓
- I fixed the debugging problems.✓
- I completed the Shopping Cart activity.✓
- I completed the Duplicate Removal activity.✓
- I explained the concepts in my own words✓.
- My notebook runs without major errors.✓
- My filename is correct.✓

13. After This Assignment, You Should Be Able To

- Create and use lists, tuples, and sets.
- Access values using indexes.

- Use slicing.
- Use common collection methods.
- Understand mutability and immutability.
- Remove duplicates using a set.
- Choose the right collection for a simple problem.

Key Principle: Don't just memorize syntax. Understand when and why to use each collection.